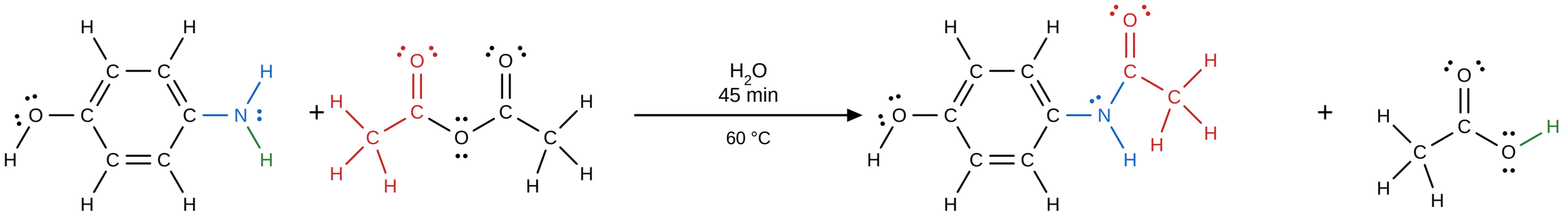

4-aminophenol + acetic anhydride → acetaminophen + acetic acid

acetaminophen

crude yield: 102.2%

recrystallized yield: 40.7%

Color key: red = acetyl group transferred from acetic anhydride; blue = amine nitrogen (becomes the amide N); green = N–H hydrogen that ends up on acetic acid.